

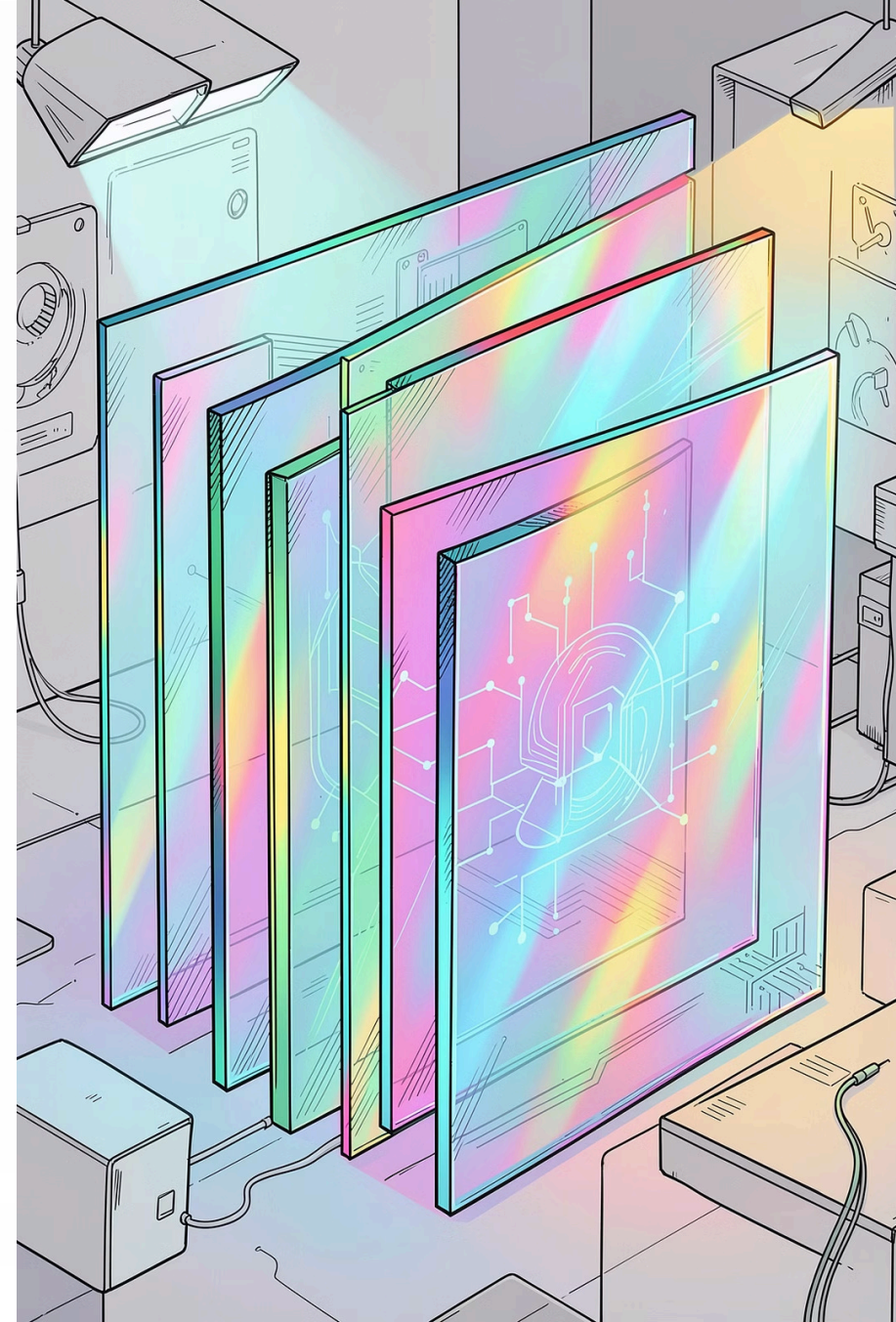
Principles of Stack Design & Mayer's Multimedia Learning

Week 12 · Lectures 23–24 · Quiz 3

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QUIZ 3



Week 12 Overview

This week bridges two powerful frameworks for designing effective multimedia learning experiences: **Stack Design Principles** and **Mayer's Multimedia Learning Principles**. Together, they provide a comprehensive toolkit for creating instructional materials that respect how the human mind processes information. Understanding both frameworks — and how they relate — is essential for anyone designing digital learning environments, educational presentations, or interactive media.

01

Lecture 23 – Principles of Stack Design

Introduction to Stack Design as a framework for organizing multimedia content. Covers the foundational principles that govern how visual and textual elements should be arranged, sequenced, and presented to optimize learner comprehension and retention.

03

Connecting the Frameworks

Examination of the relationships and overlaps between Stack Design and Mayer's principles. Identifies where the two frameworks reinforce each other and how they can be applied together in practice.

02

Lecture 24 – Mayer's Multimedia Learning Principles

Deep dive into Richard Mayer's evidence-based principles derived from decades of cognitive science research. Explores each principle in detail, including the theoretical foundations in cognitive load theory and dual coding theory.

04

Quiz 3

Assessment covering both Stack Design principles and Mayer's Multimedia Learning principles, testing understanding of key concepts, applications, and the relationship between the two frameworks.

What Is Stack Design?

Stack Design is an instructional design framework focused on the **organization and presentation of multimedia content** in layered, sequential formats. The term "stack" refers to the deliberate layering of visual, textual, and interactive elements in a way that guides the learner through information in a structured, digestible manner. Rather than presenting all content simultaneously, Stack Design advocates for progressive disclosure — revealing information in controlled sequences that reduce cognitive overload and support deeper understanding.

Core Philosophy

Information should be presented in **manageable chunks** that build upon one another. Each "layer" of the stack adds meaning without overwhelming the learner's working memory. This approach mirrors how the brain naturally processes and integrates new knowledge with existing schemas.

Key Design Concerns

Stack Design addresses questions of **spatial arrangement** (where elements appear on screen), **temporal sequencing** (when elements are revealed), and **modality balance** (how visual and verbal information are combined). These concerns directly parallel the challenges Mayer's research addresses.

Application Context

Stack Design principles are applied in **e-learning modules, slide-based presentations, interactive tutorials, and digital textbooks** — any medium where multimedia content must be structured for optimal learning outcomes. The framework is especially relevant in digital and online learning environments.

Core Principles of Stack Design

Stack Design rests on several foundational principles that govern how multimedia content should be structured. While specific formulations may vary, the following principles represent the core ideas commonly associated with this framework in instructional design contexts:

Chunking

Break complex content into **small, meaningful units**. Each chunk should represent a single concept or step, making it easier for learners to process and encode into long-term memory. Avoid presenting large blocks of undifferentiated information.

Sequencing

Arrange chunks in a **logical, progressive order** that builds understanding step by step. Earlier layers establish foundations that later layers expand upon. Sequence should follow the natural flow of reasoning or the structure of the task being taught.

Layering

Use **progressive disclosure** to reveal information gradually. Not all content needs to be visible at once — revealing layers on demand or in sequence keeps learners focused and reduces the risk of split attention across competing elements.

Alignment

Maintain **visual and conceptual consistency** across layers. Related elements should be spatially aligned, and the design language (typography, color, iconography) should remain coherent throughout the stack to support recognition and reduce extraneous processing.

These principles collectively aim to **reduce extraneous cognitive load** while supporting the learner's ability to select, organize, and integrate new information — the three cognitive processes central to meaningful learning.

Introduction to Mayer's Multimedia Learning Principles

Richard E. Mayer is one of the most influential researchers in educational psychology. His work on **multimedia learning** is grounded in rigorous experimental studies and is informed by three foundational cognitive theories. Mayer's principles have been tested across hundreds of studies and are widely regarded as evidence-based guidelines for designing effective multimedia instruction.

Dual Coding Theory (Paivio)

Humans process visual and verbal information through **separate channels**. Combining both channels effectively — rather than relying on just one — leads to deeper learning because it creates multiple pathways for encoding and retrieving information.

Cognitive Load Theory (Sweller)

Working memory has **limited capacity**. Instructional design must account for this limitation by minimizing extraneous load (unnecessary processing), managing intrinsic load (complexity of the material), and optimizing germane load (processing that contributes to learning).

Active Learning Theory

Meaningful learning occurs when learners **actively engage** in selecting relevant information, organizing it into coherent structures, and integrating it with prior knowledge. Multimedia design should facilitate these three cognitive processes rather than passively presenting content.

Mayer's Principles: Words, Images & Coherence

The first cluster of Mayer's principles addresses the **selection and combination of content elements** — what to include, what to exclude, and how words and pictures should work together.



Multimedia Principle

People learn better from words and pictures than from words alone. This is the foundational principle of multimedia learning. Combining visual and verbal representations creates dual encoding opportunities, leading to stronger memory traces and better transfer of learning.



Coherence Principle

Exclude extraneous words, pictures, and sounds. Interesting but irrelevant material competes for limited cognitive resources. Decorative graphics, entertaining anecdotes, and background music that do not directly support the learning objective should be eliminated to reduce extraneous cognitive load.



Signaling Principle

Highlight essential material with cues. Use visual or verbal signals — such as arrows, bold text, headings, or spoken emphasis — to direct learner attention to the most important information. Signaling reduces the search effort required to identify key content.



Redundancy Principle

Avoid presenting identical text and narration simultaneously. When graphics are accompanied by both on-screen text and spoken narration saying the same thing, learners experience split attention between reading and listening. Use narration alone with graphics, or text alone — not both redundantly.

Mayer's Principles: Spatial, Temporal & Modality

The second cluster of principles addresses **how and when** corresponding words and visuals should be presented to maximize integration in the learner's mind.

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Spatial Contiguity Principle Place corresponding words and pictures near each other on the screen. When text labels are far from the graphics they describe, learners must mentally search and hold information in working memory while making connections. Integrating labels directly onto or adjacent to visuals eliminates this split-attention effect and frees cognitive resources for learning.	Temporal Contiguity Principle Present corresponding narration and animation simultaneously rather than successively. When visual and verbal explanations are separated in time, learners must hold one in memory while processing the other. Simultaneous presentation allows both channels to work in concert, supporting immediate integration of the two representations.	Modality Principle Use narration rather than on-screen text to explain graphics. Because visual and auditory channels have separate processing capacities, presenting graphics with spoken narration distributes cognitive load across both channels. Using on-screen text with graphics overloads the visual channel, impairing learning — especially for complex or fast-paced material.

📌 **Key Insight:** The Spatial Contiguity, Temporal Contiguity, and Modality principles all address the same underlying problem — **split attention**. When learners must divide their focus between separate sources of information, learning suffers. These principles provide design solutions to keep corresponding information unified in space, time, and modality.

Mayer's Principles: Segmenting, Pre-training & Personalization

The third cluster of principles addresses **learner pacing, prior knowledge, and social engagement** — factors that influence how learners interact with multimedia content on a personal level.



Segmenting Principle

Break lessons into learner-paced segments. Complex content should be divided into meaningful chunks that learners can process at their own speed before moving on. Segmenting gives learners control over the pace of instruction, allowing time for the cognitive processes of selection, organization, and integration to occur fully before new information is introduced.



Pre-training Principle

Ensure learners know the names and characteristics of key concepts before the main lesson. When learners encounter unfamiliar terminology within a complex explanation, their working memory is consumed by decoding vocabulary rather than understanding relationships. Brief pre-training on key terms and components frees cognitive resources for deeper processing during the main instruction.



Personalization Principle

Use conversational style rather than formal academic language. Research shows that learners engage more deeply and perform better on transfer tests when instruction uses a first-person, conversational tone ("I'll show you...") rather than a formal, third-person style. Conversational language triggers social responses that increase learner effort and attention.



Embodiment Principle

Include visible instructors or agents that gesture and make eye contact. When learners see a human or animated agent presenting content with natural gestures and gaze cues, they experience a stronger sense of social presence. This social engagement increases motivation and cognitive effort, leading to better learning outcomes — particularly for novice learners.

Connecting Stack Design & Mayer's Principles

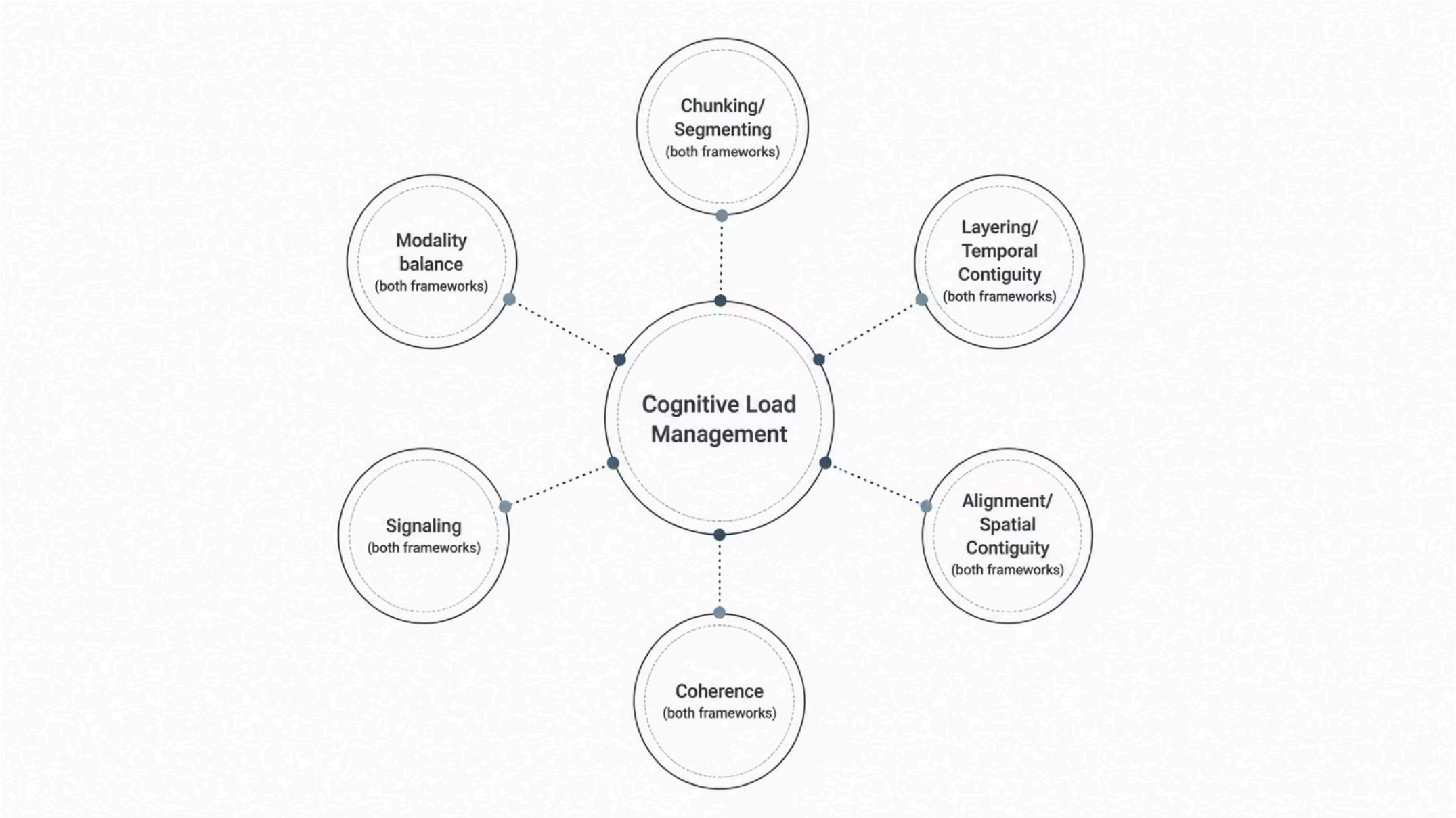
Stack Design and Mayer's Multimedia Learning Principles are **deeply complementary frameworks** that share the same cognitive science foundations. Both are concerned with managing cognitive load, supporting dual-channel processing, and facilitating active learning. Understanding their relationship allows designers to apply both frameworks with greater intentionality and consistency.

Stack Design → Mayer's Principles

Stack Design's emphasis on **chunking and sequencing** directly maps onto Mayer's **Segmenting Principle** — both advocate for breaking content into learner-controlled units. Stack Design's **layering** approach mirrors the **Temporal Contiguity Principle**, ensuring that related information is revealed in coordinated sequences rather than all at once. The **alignment** principle in Stack Design reflects Mayer's **Spatial Contiguity Principle**, keeping related visual and textual elements in close proximity.

Mayer's Principles → Stack Design

Mayer's **Coherence Principle** reinforces Stack Design's goal of eliminating extraneous content from each layer of the stack. The **Signaling Principle** informs how Stack Design should use visual hierarchy and cues to guide learners through each layer. Mayer's **Modality Principle** provides specific guidance for Stack Design decisions about whether to use narration, on-screen text, or a combination when presenting layered content.



Key Takeaways & Quiz 3 Preparation

As you prepare for Quiz 3, focus on understanding not just the individual principles but the **cognitive reasoning behind each one**. The principles are not arbitrary rules — they are grounded in how human memory and attention work. Being able to explain *why* a principle works will help you apply it flexibly in new design situations.

→ Stack Design Core Ideas

Remember the four pillars: **Chunking** (break content into meaningful units), **Sequencing** (arrange in logical order), **Layering** (progressive disclosure), and **Alignment** (visual and conceptual consistency). These principles reduce extraneous load and guide learners through complex material systematically.

→ Theoretical Foundations

Be prepared to explain how **Dual Coding Theory**, **Cognitive Load Theory**, and **Active Learning Theory** underpin both frameworks. Understand the distinction between **extraneous, intrinsic, and germane cognitive load**, and how good design minimizes the first while supporting the third.

→ Mayer's 12 Principles – Grouped

Organize Mayer's principles into three clusters for easier recall: **Content Selection** (Multimedia, Coherence, Signaling, Redundancy), **Presentation Timing & Placement** (Spatial Contiguity, Temporal Contiguity, Modality), and **Learner Engagement** (Segmenting, Pre-training, Personalization, Embodiment). Know the definition and cognitive rationale for each.

→ Framework Relationships

Understand how Stack Design principles **map onto and reinforce** Mayer's principles. Both frameworks share the goal of reducing cognitive overload and supporting the three cognitive processes of **selecting, organizing, and integrating** information. Be ready to give specific examples of how the two frameworks complement each other in practice.

📌 **Quiz Tip:** Questions may ask you to identify which principle is violated in a given design scenario, or to recommend design changes based on specific principles. Practice applying each principle to real multimedia examples — this will prepare you for both recall and application questions on Quiz 3.